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Examining the Effectiveness of Interventions to Reduce Discriminatory Behavior at Work: An Attitude Dimension Consistency Perspective

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Academic interest in reducing discrimination has produced substantial research testing interventions to mitigate biased outcomes. However, disparate findings and a scarcity of studies examining work-related behavioral measures make it challenging to determine which interventions are better suited to reduce workplace discrimination. Derived from the tripartite theory of attitudes and the principle of compatibility, I develop a conceptual model mapping the attitude focus of interventions and code studies in this literature from the past two decades for these common properties. Based on a meta-analysis of 70 articles totaling 208 effect sizes, I test this conceptual model, finding that it helps explain why some interventions to reduce discrimination yield superior outcomes relative to others. In particular, results indicate that passive interventions, such as short-term education or reminders of bias processes, are largely ineffective in shifting behavior. Conversely, the class of interventions that targets behavior directly by attempting to inhibit the manifestation of bias (e.g., making individuals accountable for their decisions or changing social norms) emerged as the most helpful category of interventions in this area. Overall, results support a key prediction of the attitude dimension consistency perspective, demonstrating that aligning the attitude dimension primarily targeted by an intervention and the outcome measured could lead to improved results in this area.

Keywords: discrimination; diversity, equity, and inclusion; interventions; prejudice; bias reduction

In recent years, increased public concern over social justice issues led corporate leaders worldwide to prioritize diversity, equity, and inclusion (DEI) policies in their organizations. Surveys conducted a few months to a year after George Floyd's death showed that DEI initiatives remained a top concern among C-suite executives, prompting several pro-diversity and equity policy announcements from organizations across industries (Colletta, 2022; Fortune-Deloitte, 2020; Glazer & Francis, 2021; Ryan & Fiatte, 2022). While some of these initiatives ramped up current standard organizational practices in this area, such as increasing diversity training, others introduced more novel strategies, such as implementing anonymized resumes (Agovino, 2020; Carlos, 2021). A key open question to management practice that should be informed by theory and grounded in empirical evidence is which initiatives companies should adopt based on their propensity to produce meaningful change in organizations. While much research has been conducted to address these issues, traditionally, this has been a challenging question to answer, as interventions have met with mixed success in this literature in two distinct ways. First, interventions that appear effective under certain conditions produce null or weaker effects in different contexts (Aboud et al., 2012; Hsieh et al., 2022; Rodon & Franco-Guillén, 2014). Despite this, only a small proportion of research examining interventions to decrease bias investigates work-related outcomes, with most interventional studies assessing more general, nonwork-related attitudes and behaviors (Paluck et al., 2021). This is notable since workplace settings could potentially influence the effectiveness of interventions in different ways. For example, research has shown that adopting a business framing to decision making can influence the cognitive processing of information (Rees et al., 2022). In addition, as certain work-specific interventions can be examined in a way that is distinct from other contexts (e.g., anonymizing resumes, reminding individuals about equal employment opportunity policies and laws), an evaluative review of interventions centered only on workplace outcomes is a pressing need.

Second, in many cases, an intervention might effectively reduce one aspect of bias (e.g., discriminatory behavior) but not another (e.g., prejudicial evaluations; Forscher et al., 2019; Paluck et al., 2021). In this regard, I argue that these results occur because different interventions likely target distinct *dimensions* of attitudes. Current scholarship lacks a framework that systematically classifies which element of attitudes is primarily altered by which interventional approach, hindering researchers' ability to predict how individuals are likely to respond to these various interventions. However, understanding the interplay between an intervention's attitudinal focus and the desired outcome is critical for building better interventions in this area. To address this limitation, I review studies in this literature, mapping intervention approaches to the primary attitude dimension affected by each intervention procedure, uncovering commonalities in how they have been typically implemented. In developing this

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intervention classification system, I leverage the tripartite model of attitudes (Rosenberg & Hovland, 1960), which identifies the distinct components that underpin attitudes toward social groups. The resulting taxonomy offers a comprehensive view of which aspect of attitudes the different intervention strategies tested so far primarily intervene on. Using this categorization, I formulate and test a framework showing which class of interventions works best to reduce discriminatory behavior in the workplace and why.

Overall, this examination expands the cumulative knowledge in this literature in three ways. First, a taxonomy that methodically identifies intervention categories based on attitude dimensions should enhance the predictive ability of interventional procedures—helping improve the efficacy of future intervention approaches. Second, by demonstrating that differences exist in effectiveness to reduce biased outcomes based on which attitude component is targeted by different interventions, this work contributes a novel perspective to the stereotypes and discrimination scholarship by clarifying one reason *why* certain interventions produce more successful results than others. Last, by identifying the most effective intervention approaches aimed at reducing workplace discrimination specifically, this work offers guidance to leaders and practitioners in their efforts to combat one of the most critical and persistent societal challenges facing organizations (Amis et al., 2020; Sharma & Mann, 2018; Yang & Liu, 2021).

Theoretical Background and Conceptual Model

Attitude Dimensions

Bias is an unfair response (evaluative, emotional, cognitive, or behavioral) that favors or disfavors an individual or the group to which they belong over others (Dovidio & Gaertner, 2010). One way bias, or attitudinal preferences, are broadly distinguished in the literature is through the tripartite model of attitudes (Rosenberg & Hovland, 1960). According to the contemporary view of this theoretical model, attitudes reflect an appraisal of a target associated with three dimensions: cognitive (thoughts, ideas, or beliefs), affective (negative or positive emotions), and behavioral (overt actions and responses; Fabrigar et al., 2005; Kaiser & Wilson, 2019). When applying this model to intergroup attitudes, these manifestations relate to the following concepts: stereotypes constitute the cognitive dimension, representing socially learned mental schemas of a given group; prejudice is the affective dimension characterized by an overall positive or negative valence; and discrimination embodies the behavioral dimension, capturing attitudes manifested in actions and decisions (Esses & Dovidio, 2002; Fiske, 1998).

Due to growing awareness and interest in counteracting bias, much research has focused on understanding and testing interventions to reduce its detrimental effects. The impact of these approaches has produced inconsistent results, however, with some interventions showing improvements across all attitude dimensions while others changing certain aspects of attitudes but not others. For instance, Turner and West (2012) found that imagining contact with a stigmatized group member helped reduce all dimensions of bias: Participants reported improved beliefs about the character of the target, more positive feelings, and demonstrated more approachable behavior (i.e., choosing to sit closer to the partner). Similarly, Nier et al. (2001) showed that invoking a common identity increased both positive affective responses and prosocial behavior toward Black confederates. Conversely, Gloor and Puhl (2016) found a perspective-taking intervention increased empathy and improved affective reactions toward obese individuals while having no impact on behavioral intentions relating to social interactions (e.g., working in the same office or socializing at an event). Meanwhile, Paluck (2011) evaluated the effect of a peer influence intervention on several bias outcome measures, finding that the intervention was effective in changing behavior in the treated group but not their beliefs. Likewise, field experiments leveraging intergroup contact interventions by Mousa (2020) and Scacco and Warren (2018) reduced discriminatory behavior but showed no effects or inconsistent results on prejudicial attitudes.

The mixed results in the bias intervention literature stem from various factors, such as how constructs are measured or idiosyncratic aspects of intervention paradigms across studies (Dixon et al., 2010; Gloor & Puhl, 2016; Paluck & Green, 2009; Skinner & Meltzoff, 2019). Yet, given that a number of studies measuring different attitudinal dimensions in the same investigation show results where the intervention is successful on one dimension but produces null or more modest effects on the others, suggests another possible source of this inconsistency: that interventions are fundamentally targeting different dimensions of attitudes (Álvarez-Delgado et al., 2022; Hanisch et al., 2016). In this case, while an intervention can also change other attitudinal dimensions, stronger effects should occur when there is concordance between the intervention's focal attitude dimension and the outcome measured (Chambon et al., 2022; Kok et al., 2016). Therefore, to improve theoretical and practical understanding in this area, it is helpful to consider the focal attitudinal dimension of the intervention relative to their effectiveness in changing workplace bias measures.

The tripartite model of attitudes has been employed in various settings beyond the workplace to examine bias reduction interventions. For instance, researchers have examined reducing different attitudinal measures of political bias, such as countering biased cognitive and emotional responses rooted in political affiliation by providing participants with different viewpoints or empathetic narratives (Mernyk et al., 2022; Voelkel et al., 2023). Similarly, in health research, the tripartite framework has informed interventions to reduce mental health stigma and bias. These investigations leveraged education and awareness campaigns to target cognitive components, personal narratives to modify emotional reactions, and supportive behaviors and language to promote behavioral change (Dalky, 2012; Na et al., 2022). Most prominently, consumer psychology research has shown that the consistency of affective and cognitive-based messages with the desired outcome can enhance their persuasiveness (Clarkson et al., 2011; Fabrigar & Petty, 1999). Essentially, while psychological interventions have employed the tripartite model in different research areas, this article's focus on matching intervention approaches with outcome measures in the context of workplace bias offers a novel perspective for addressing intergroup bias more effectively.

Taxonomy of Interventions

I ground the taxonomy of interventions on the tripartite model of attitudes, as it describes how different components of attitudes (i.e., cognitive, affective, and behavioral) serve as sources of information from which individuals develop summary evaluations about others—and could, therefore, be individually targeted by interventions. In addition, I differentiate between active interventions, which employ techniques that target the different attitudinal components, and passive interventions, which only leverage the disclosure of information, such as in messages or leaflets (Albarracín et al., 2005; Kuningas et al., 2020). Passive interventions do not attempt to change any specific attitude component and, therefore, completely rely on the evaluator's internal motivation to update their evaluations or behavior. Figure 1 depicts this conceptual model mapping intervention categories to attitude dimensions.

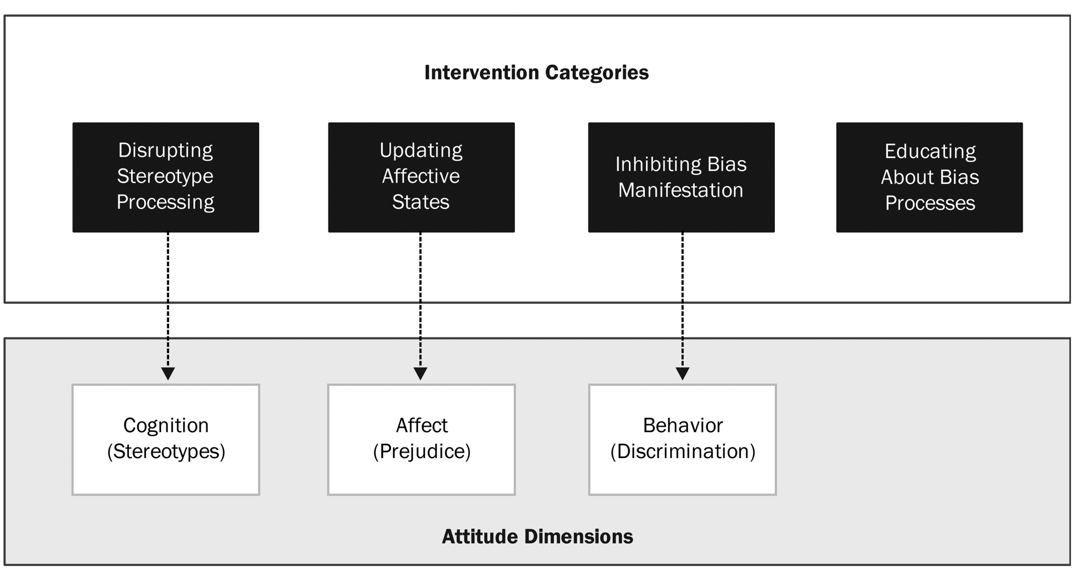
In an effort to categorize interventions in the bias literature, Lenton et al. (2009) offered three types grounded on stereotyping: inhibiting stereotype activation, emphasizing stereotype heterogeneity, and preventing stereotype expression. Previously, Blair (2002) put forward that stereotypes and prejudice are malleable and can be influenced by factors such as one's motives, specific strategies, perceiver's focus of attention, and contextual cues. Building on these frameworks, I propose four categories that organize interventions according to which attitude dimension the procedure primarily operates (i.e., cognitive, affective, behavioral, or none). To determine each intervention's attitude dimensional focus, I conducted a review of literature identifying the key mechanism underlying attitude change in interventional studies and coded them accordingly. Table 1 defines each intervention type and describes the empirical and theoretical reasoning for sorting intervention types into each category. Table 2 provides a summary of the key features of each intervention category. This analysis resulted in the following bias reduction intervention categories:

- Disrupting stereotype processing (active interventions operating primarily on cognition).
- Updating affective states (active interventions operating primarily on affect).
- Inhibiting bias manifestation (active interventions operating primarily on behavior).
- Educating about bias processes (passive interventions, with no primary attitude dimension targeted).

Disrupting Stereotype Processing

This class of interventions mainly operates on the cognitive dimension of attitudes and is designed to interfere with the typical processing of stereotypes. This can be accomplished by shifting attentional focus away from stereotypical information or by managing choice architecture (i.e., how alternatives are presented to a decision-maker). As a result, these interventions function by changing how information about a target is perceived or by influencing to which information the evaluator directs their attention. For example, Reynolds et al. (2021) induced participants to adopt a calculative mindset, while Uhlmann and Cohen (2005) asked participants to commit to important criteria prior to evaluations. Singletary and Hebl (2009) provided individuating information about targets, while Feng et al. (2020) manipulated the decision environment by creating partition dependence in the presentation of candidates. A subcategory of this class of interventions aims to prevent stereotype activation completely. In this approach, the strategy employed precludes stereotypes from being triggered altogether during target evaluation, so they do not factor into the decision-making process. A prominent example of this intervention is Goldin and Rouse's (2000) study examining orchestra hiring over time, which concealed candidate identities with a screen during auditions. The anonymization of candidate information is a type of choice architecture intervention aimed at eliminating the influence of demographic information in the

Figure 1
Conceptual Model of Bias Reduction Interventions and Their Corresponding Attitude Dimensional Focus



Note. This model illustrates the attitude dimension that is the *primary* target of the four intervention categories identified in this review. The present model proposes that, while interventions may change dimensions that are not the primary focus of interventions, in general, stronger effects should be achieved when there is correspondence between the attitudinal focus of an intervention and the measured outcome.

Table 1
Intervention Types: Descriptions and Basis for Classification

Intervention type	Description	Basis for classification
Category: Disrupting stereotype processing		
Blinding/anonymizing resume	Removal or masking of demographic-related information (e.g., gender, race) from a candidate's job application materials	Suppresses the access to social category signals during evaluations (Lacroux & Martin-Lacroux, 2020)
Calculative mindset	Inducing evaluators to adopt a quantitative or rational frame of mind when analyzing qualitative problems and situations	Adoption of a calculative mindset shifts information processing to be more data-driven, detached, impersonal, and less influenced by affective responses (Reynolds et al., 2021).
Countersterotypic information	Disclosing information about the target that directly contradicts the stereotypes to which the target belongs	Surprise has been found to mediate the reduction of intergroup bias in counterstereotype interventions. When the target cannot be easily classified, it triggers the perceiver's need for information seeking to resolve the discrepancy, with this process altering stereotypic representations of the target (Hutter & Crisp, 2005; Hutter et al., 2009; Kunda et al., 1990; Maguire et al., 2011; Prati et al., 2015).
Individuating information	Providing information about an individual that is unrelated to their membership in a group (i.e., not linked to existing associative information, but also not contradicting these); inducing evaluators to think about targets as individuals rather than as part of a group.	Reduces the reliance on stereotypes, favoring attention to the individuating information in evaluations (Ruggs et al., 2011; Singletary & Hebl, 2009)
Isolated versus collective decision	Evaluators select group members for multiple slots in a decision task (collective decision) rather than selecting members for a single slot or role (isolated choice).	In collective decision settings, information about the diversity of choices becomes more salient to evaluators (Chang et al., 2020).
Joint evaluation versus separate evaluation	During evaluation, two or more options (e.g., job candidates) are presented concurrently rather than one at a time.	Simultaneous evaluation of candidates shifts focus from group stereotypes to individual performance (Bohnet et al., 2016).
Majority signaling	The target of an evaluation presents information about themselves in a way that conceals or downplays minority cues or highlights signals of belonging to majority groups.	Alters how evaluators perceive one's demographic information so that the target's group membership is more likely to be categorized as the majority's (Ruedin & Van Belle, 2023).
Nonstereotypic association	Evaluators are trained to associate nonstereotypical concepts (i.e., not culturally typical) with a particular group.	Training aimed at negating category associations is expected to reduce the automatic activation of stereotypes (Kawakami et al., 2005).
Partition dependence	Options are presented to evaluators partitioned into different grouping configurations.	Different partition arrangements affect how evaluators perceive information, which influences how they allocate resources. Decision-makers have a tendency to diversify their selections based on how options are partitioned (Feng et al., 2020).
Structured evaluation	Introduces standardization into the evaluation process by having evaluators utilize standardized appraisal criteria and anchored rating scales in assessing alternatives.	Directing attention to clearly defined criteria and using standard benchmarks during evaluation reduces ambiguity in the process, which decreases reliance on stereotypes and heuristics in decision making. (Campion et al., 1997; Kutcher & Bragger, 2004).
Category: Educating about bias processes		
Diversity training	Evaluators receive information (e.g., via multimedia online or in-person workshops) to raise awareness of bias processes and their negative consequences. In some instances, diversity training might also include taking the Implicit Association Test and learning strategies to combat bias.	No primary attitude component targeted. No alterations of cognitive, affective, or behavioral dimensions occur by solely raising awareness of bias without a motivational frame and skills development present (Bezrukova et al., 2016; Burns & Monteith, 2019). Diversity training and bringing attention to stereotypes and bias can also backfire (Pendry et al., 2007; Rawski & Conroy, 2020; Robinson et al., 2020).
Egalitarian message	Evaluators read a nondiscrimination statement or policy (e.g., equal opportunity employment message) and are instructed to base their conclusions solely on job-relevant factors.	No primary attitude component targeted. Equal opportunity and egalitarian messages are passive policies that do not require the evaluator's involvement nor compel their compliance (Crosby et al., 2006).

(*table continues*)

Table 1 (*continued*)

Intervention type	Description	Basis for classification
Category: Inhibiting bias manifestation		
Accountability	Evaluators are informed that they will need to explain or justify their ratings, decisions, or conclusions.	The expectation that one might be asked to justify one's action is recognized as a causal factor of behavior in social psychology (Frink & Klimoski, 2004; Hall et al., 2009). This occurs because when decision-makers feel accountable to others, they scrutinize their own behavior, relying more on evidence than stereotypes (Dobbin & Kalev, 2017). Accountability theory explains that this occurs due to self-presentation motivations (conforming to behavior that is perceived as desirable prevents critical reactions, preserves, or improves reputation; Dobbs & Crano, 2001; Frink & Ferris, 1998; Lerner & Tetlock, 1999).
Affirmative action	Evaluators are informed that when they make candidate selections, they must meet a certain group composition benchmark (i.e., some representation standard) and that this will be enforced in some fashion.	Policies and laws strongly influence behavior by serving as a deterrent due to legitimacy, easing coordination, and signaling information about risk and public attitudes (McAdams, 2015; Tyran & Feld, 2002).
Social norms	Disclosing information on the socially acceptable and appropriate standard of conduct in a particular social setting to evaluators.	Normative information is known to guide behavior in social situations and to be a vehicle of behavioral change due to the negative social consequences of nonconformity (Prentice & Paluck, 2020; Tankard & Paluck, 2016).
Category: Updating affective states		
Acknowledgment	The target highlights the stigmatized aspect of their identity when communicating with the evaluator.	When the target of an evaluation acknowledges their stigmatized identity, they elicit less negative feelings such as disdain, pity (Davis, 1961). They are perceived as well-adjusted, which eases the interactions by reducing tension and discomfort during the interaction (Hebl & Skorinko, 2005; Hebl et al., 2000). This intervention reduces discrimination via the evaluator's internal motivation to reduce prejudice (Kirgios et al., 2022) induced through negative self-directed affect such as guilt and shame (see self-regulation of prejudice model; Monteith et al., 2010).
Common identity	Emphasizing the target's existing superordinate or shared group membership with the evaluator	The common ingroup identity model proposes that recategorization to a common superordinate group extends positive ingroup feelings to former outgroup individuals (Capozza et al., 2010; Gaertner et al., 1993). Research shows this occurs due to a reduction in intergroup threat and anxiety, which in turn increases outgroup positive evaluations (warmth, sympathy; Riek et al., 2010).
Imagine contact	Inducing the evaluator to imagine a positive social interaction with a member of the target group prior to evaluation	The imagined contact hypothesis suggests that positive attitudes and behavior ensue from this exercise (Crisp & Turner, 2009). Experiments demonstrate the mediating role of emotions in decreasing prejudice (i.e., decreased anxiety, increased empathy toward the outgroup), while no cognitive role is found in this intervention (Pettigrew & Tropp, 2008; Tropp & Pettigrew, 2005; Turner et al., 2013; Vezzali et al., 2013).
Multicultural experience	Introducing evaluators to aspects of a culture different from their own, such as in pictures, music, and movie clips (e.g., multimedia presentation or training)	Multicultural experience reduces prejudice via contact (primarily mediated by emotional factors such as intergroup anxiety; Binder et al., 2009; Tadmor et al., 2012) and common identity effects of a superordinate nature (i.e., via a stronger identification with all of humanity; Sparkman & Eidelman, 2018; Sparkman & Hamer, 2020).

(*table continues*)

Table 1 (*continued*)

Intervention type	Description	Basis for classification
Perspective taking	Prompting evaluators to imagine how the target feels or the ways that the target's identity affects their life	Prejudice reduction with this intervention is mediated by empathy and self-other overlap, a measure of emotional closeness. Perspective taking increases liking, sympathy, intention to help, and psychological closeness with the outgroup (Ku et al., 2015; Myers & Hodges, 2013; Shih et al., 2009; Todd & Galinsky, 2014; Vescio et al., 2003; Waugh & Fredrickson, 2006)
Warmth/positivity	Increasing signals of a warm and friendly personality (when this cue is not directly countering a group stereotype of "coldness"). Targets are instructed to behave positively during an evaluation (i.e., upbeat, smiling frequently, being pleasant) or are described as affable and sociable.	Targets who display positivity are rated higher in likability and social adeptness (Ruggs et al., 2011). The warmth dimension is associated with liking in evaluative judgments; Fiske et al., 2007).

evaluation process, forcing the decision-maker to concentrate their appraisal solely on human capital factors instead. In their varied ways, this class of interventions diverts focus to other cues or re-arranges the decision environment to compete with or overwhelm stereotype processing. Thus, in this category, the mental schemas or stereotypes about the target are largely unaffected—rather, the objective is to interfere with their regular processing.

Updating Affective States

In this category, interventions aim to index new affective states that are inconsistent with previous negative group categorizations. Alternatively, they can involve an emotional appeal that gives rise to a cognitive dissonant state to facilitate behavior change (e.g., via guilt or shame). Examples include imagining contact with an individual from the target group (Moss-Racusin & Rabasco, 2018), evoking a shared identity between the target and the decision-maker (Schmader et al., 2013), and taking the perspective of the target of evaluation (Mendoza et al., 2019). Many of these interventions act by reducing negative emotional states, such as intergroup anxiety, or increasing positive feelings, such as empathy toward the social group or evaluation target. Accordingly, decreased anxiety and increased empathy mediated the effectiveness of intergroup contact interventions, while increasing knowledge of the group, a more cognitive-based approach, did not (Pettigrew & Tropp, 2008). Importantly, this class may not completely "update" impressions of the group as a whole. Instead, they can initiate a subcategorization process where the affective information perceived to be uncommonly associated with the original group is appended to a subcategory of the target group. This allows the feelings and beliefs associated with the subcategory to be ascribed to the target of evaluation (Williams, 2001). In essence, these interventions operate

Table 2
Taxonomy of Bias Reduction Interventions

Intervention category	Primary attitude dimension targeted by intervention	Key defining characteristic	Intervention approach example
Disrupting stereotype processing	Cognitive (interferes with or prevents processing of stereotypes)	• Shifts attention away from stereotype to other information • Blocks activation of stereotype • Does not attempt to alter the content of attitudes	• Structured evaluation • Individuating information • Anonymizing resumes
Updating affective states	Affect (indexes new affective states that are inconsistent with previous categorizations or causes a dissonant state)	• Increases positive feelings, decreases negative feelings toward the target or the self • Elicits an emotional response to adjust one's behavior • Could begin a process of altering the content of attitudes via affect	• Perspective taking • Imagining contact • Common identity
Inhibiting bias manifestation	Behavior (motivates individuals not to act on stereotypes or bias)	• Offers a strong external motivational cue to conform with a certain behavior • Involves some level of cost/sanctions to the individual • Does not attempt to alter the content of attitudes	• Social norms • Accountability • Affirmative action
Education about bias processes	None (creates awareness, reminds individuals of stereotypes and bias)	• Directs attention to bias and stereotypes processes • Attempts to alter the content of attitudes in short-term training/awareness interventions • Relies heavily on individuals' internal motivation to not act on stereotypes and prejudices	• Diversity training • Equal employment opportunity message

on emotions so that the evaluator feels less negative or more positive toward the target, the self, or the situation than in the baseline condition.

Inhibiting Bias Manifestation

This intervention category provides strong external motivational cues for individuals to not act on stereotypes and prejudice, directly targeting discriminatory behavior. This is typically accomplished via normative or regulatory aspects in the social environment that exert pressure on the individual to conform to a standard of conduct. Therefore, this class of interventions changes, guides, and informs behavior by the presence of an implied or explicit negative sanction resulting from noncompliance with a certain behavior. Examples of this intervention category include accountability, social norms, and affirmative action policies. For instance, Nadler et al. (2014) tested accountability by asking participants to explain their ratings to an authority figure. Duguid and Thomas-Hunt (2015) informed participants that the normative behavior in that context was to try to overcome biased preconceptions when evaluating others.

Educating About Bias Processes

This category of interventions is largely informational, aimed at raising awareness or reminding individuals about bias and its negative consequences. Rather than specifically targeting an attitude dimension, passive interventions rely solely on the individual's internal motivation to change attitudes or behavior. Examples include diversity training, where individuals learn about stereotypes and bias processes, are instructed to be vigilant, and may receive guidance on avoiding discriminatory behavior (Chang et al., 2019; Devine et al., 2012, 2017). Alternatively, individuals are reminded about egalitarian policies to attend to prior to an evaluation (Lindner et al., 2014; Nadler et al., 2014). This class of interventions is the only one that attempts to combat biased outcomes by bringing stereotyping and bias processes into explicit view, contrary to the other methods, which do not mention or overtly raise them as issues to be considered by individuals. As a result, they carry the greatest risk of backfiring (Atewologun et al., 2018; Hideg & Wilson, 2020).

Workplace Behavior Dimension

The classification system outlined previously seeks to inform and improve predictions on the most effective intervention *categories* to reduce the different forms of biased outcomes in organizations. In addition, this study aims to identify which intervention *types* effectively reduce workplace discrimination. Because it remains unclear from a theoretical standpoint which intervention approaches (rather than broad categories) will produce the best results in reducing discriminatory behavior, I approach this critical question on an exploratory basis.

> *Research Question 1:* Which intervention types are most effective in reducing discriminatory outcomes in the workplace?

In the organizational literature, discriminatory behavior is further distinguished between formal and informal discrimination (Chung, 2001; Levine & Leonard, 1984). Formal discrimination relates to the biased allocation of scarce organizational resources (e.g., hiring, promotion, salary allocations), while informal discrimination captures the quality of interactions among employees (e.g., hostility, ostracism; Welle & Heilman, 2007). In other words, formal discrimination depicts behavior typically considered illegal in the workplace. In contrast, informal discrimination does not entail unlawful behavior but still reflects animus toward an individual based on their group membership.

Notably, evidence from field studies suggests that interventions can produce distinct patterns for these two forms of discrimination. For instance, some field studies show no differences in formal discrimination measures before interventions (e.g., pregnant vs. nonpregnant confederates are allowed to complete job applications at the same rates) or following interventions (e.g., no callback rate changes for stigmatized groups across the treatment and control arms of the intervention). On the other hand, these studies demonstrate noticeable effects of the same interventions on informal discrimination, such as reductions in negativity or rudeness and increases in friendly behavior toward stigmatized groups (Barron et al., 2011; King & Ahmad, 2010; Singletary & Hebl, 2009). This discrepancy could occur due to the reality that formal discrimination, being typically illegal, is less likely to be openly practiced and, thus, harder to detect and measure in the field. This implies that while formal discrimination may remain relatively static due to legal constraints, informal discrimination, driven by more discretionary behaviors, is more amenable to change through interventions. As the differential impact of interventions on formal and informal discrimination is a relatively unexplored area of study, I take an exploratory approach to this question:

> *Research Question 2:* Are interventions targeting the behavioral dimension of attitudes more effective in reducing informal as opposed to formal discriminatory outcomes?

The (In)Consistency of the Attitude–Behavior Relationship

While research on attitudes spans concepts like structure, measurement, content, and formation theories, the relationship between attitude (here, referring to its evaluative manifestations only: cognitive and affective dimensions) and ensuing behavior features as a prominent topic in social psychology (Eagly & Chaiken, 1993). Early studies and Wicker's (1969) seminal review shattered previous assumptions that evaluative attitudes are closely related to overt actions—firmly establishing the concept of attitude–behavior inconsistency. Although later research demonstrated that fairly strong attitude–behavior correlations can occur under certain circumstances (e.g., voting, sports attendance), the notion of an overarching cross-situational relationship between evaluative attitudes and behavior has not been supported (Zanna et al., 1980). The absence of a reliably consistent association between evaluative attitudes and behaviors is especially critical in prejudice research, where social desirability concerns and situational forces exert an outsized influence in the poor attitude–behavior correspondence (Greenwald et al., 2009; Phillips & Olson, 2014). Empirical work in this area suggests that while correlated, there is clear evidence for the independence of attitude dimensions, implying limits on the conditions when interventions attempting to modify one dimension of attitudes will generate associated changes in others (Forscher et al., 2019; Paluck et al., 2021; Schütz & Six, 1996). In light of this, some interventions may produce superior results than others in reducing biased outcomes when examining interventions based on their attitude dimensional focus.

One way that evaluative attitudes and behavior show stronger associations is through the principle of compatibility (Ajzen & Fishbein, 1977). This principle posits that to maximize the association between evaluative attitudes and behavior, they should be measured at the same level of specificity with respect to target, action, context, and time (Smith & Haslam, 2017). In other words, while more general attitudes can correlate with some specific behavior, stronger attitude predictors occur when these match the behavior's level of concreteness (e.g., measuring participants' specific attitudes toward enrolling in a voluntary diversity training course this quarter rather than their general attitudes toward diversity). The principle of compatibility has received strong empirical support, generally demonstrating much stronger correlations between evaluative attitudes and behavior when this principle is followed (Ajzen, 2005). For example, Kraus (1995) meta-analyzed 88 studies investigating the relationship between evaluative attitudes and behavior, finding correlations of $r = 0.54$ under conditions of high correspondence and of $r = 0.13$ under conditions of low correspondence.

Similarly, I argue that most interventions involve modifying an underlying primary attitude dimension. While an intervention can impact other dimensions, the magnitude of its effectiveness should be stronger when there is an alignment between the dimensional focus of the intervention and the outcome variable. For instance, if an intervention primarily targets the affective component, prejudicial measures are likely to change to a greater extent compared to the other dimensions. With that said, if the researcher is measuring behavioral outcomes, the intervention's effectiveness might not emerge or be small in magnitude. Therefore, when the main attitude dimension targeted by an intervention matches the measured outcome, a relatively stronger effect is expected to occur in reducing bias measures.

> *Hypothesis 1:* Interventions that directly target the behavioral dimension of attitudes (i.e., inhibiting bias manifestation category) will be more effective when paired with behavioral outcome measures compared to nonpaired combinations.

> *Hypothesis 2:* Interventions that directly target the affective dimension of attitudes (i.e., updating affective states category) will be more effective when paired with affective outcome measures compared to nonpaired combinations.

> *Hypothesis 3:* Interventions that directly target the cognitive dimension of attitudes (i.e., disrupting stereotype processing category) will be more effective when paired with cognitive outcome measures compared to nonpaired combinations.

Passive interventions are largely informational, aimed at generating awareness or serving as reminders for a cause or issue, such as the importance of voting or vaccination. In line with these aims, passive interventions typically include posters, handouts, and other educational tools, requiring self-discipline and well-motivated individuals to be effective (Milot et al., 2015). By bringing awareness or reminding individuals about stereotypes and bias processes, this class of interventions does not attempt to disrupt or prevent the cognitive processing of categorizations; rather, it directs attention to it. This can actually lead to *more* discrimination, as it activates stereotypes, sends the message that biases are involuntary and normal, may trigger identity threat among advantaged groups, and can spark managerial resistance (Atewologun et al., 2018; Dobbin & Kalev, 2017; Hideg & Wilson, 2020; Hirsh & Cha, 2017; Osman, 2021). Likewise, while educational interventions may increase sympathy toward typically disadvantaged groups among already self-motivated, nonprejudiced decision-makers, evidence shows they tend to backfire for those who hold prejudices. In other words, these interventions tend to move attitudinal measures for individuals who are already intrinsically motivated to act in an unbiased manner while having minimal, null, or detrimental effects on those who are not as motivated to do so (Chang et al., 2019; Devine et al., 2012; Jackson et al., 2014). In addition, by emphasizing outside control, these interventions can, at times, undermine individuals' motivation to regulate their prejudice, producing a backlash effect instead (Chow et al., 2021; Legault et al., 2011; Onyeador et al., 2021). This suggests educational interventions are also generally ineffective at constructively updating affective states. Finally, because they do not carry implied or explicit sanctions for the individual to adhere to conduct standards, educational interventions do not directly target actions, thus being largely ineffectual at changing behavior. For these reasons, educational interventions should not produce substantive changes in any of the measures of bias.

> *Hypothesis 4:* Interventions that do not directly target any attitude dimension (i.e., educating about bias processes category) will be less effective in reducing workplace bias measures relative to all other intervention categories.

Method

Search Strategy

I first located relevant studies by searching APA PsycInfo, Google Scholar, Web of Science, Scopus, and ProQuest databases for a combination of keywords relating to interventions (e.g., experiment, treatment, etc.) involving workplace outcomes (e.g., recruit*, employ*, etc.) within the context of intergroup bias (e.g., prejudic*, discriminat*, etc.) In addition, I performed backward searches of previous meta-analyses and reviews on this topic to identify further records (e.g., Nandigama & Shyamsunder, 2021; Paluck et al., 2021; Stephens et al., 2020). Finally, I posted requests for unpublished studies on listservs for the Society for Personality and Social Psychology (SPSP Connect!), the Academy of Management's Organizational Behavior (OB), Diversity, Equity and Inclusion (DEI), and Managerial and Organizational Cognition (MOC) divisions. No limits were placed on publication date, with the final range of studies analyzed published between 2000 and 2022. Appendix A contains the full keyword parameters used in the literature search.

Selection Criteria

I evaluated articles in two stages. In the first stage, only the title and abstract were screened for potential inclusion in the analysis.

Specifically, if the title and abstract did not refer to a study testing an intervention related to intergroup bias, it was excluded. During the second stage, I identified potentially eligible studies based on the full-text evaluation of the article. The inclusion criteria at this stage involved: (a) examination of the effects of a bias reduction intervention[1]; (b) containing at least one attitudinal outcome measure relevant to the workplace; and (c) reporting sufficient statistics to derive standardized effect sizes. The complete selection process flowchart for this analysis is depicted in Figure 2.

Data Coding

Intervention Types

The intervention type was determined based on the description actually used in the treatment group rather than the label given by the authors. For example, a study labeling an intervention "commitment to hiring criteria" that consisted of evaluating candidates based on predetermined critical factors for a role was coded as "structured evaluation" since much of its features are consistent with this intervention approach.

Intervention Category

To determine which aspect of attitude was the focal point of a given intervention, an analysis of the description of the treatment and control conditions was performed for each study and classified according to the proposed taxonomy:

1. *Disrupting stereotype processing*: Interventions targeting the cognitive component of attitudes by attempting to disrupt the processing or retrieval of stereotypes, primarily impacting what information the individual focuses on (e.g., anonymizing resumes, structured evaluation, etc.).
2. *Updating affective states*: Interventions mainly targeting the affective component of attitudes by attempting to increase positive or decrease negative emotions toward the target of evaluation or the self (e.g., imagining contact, perspective taking, etc.).
3. *Inhibiting bias manifestation*: Interventions targeting the behavioral component of attitudes by providing a strong external motivation for individuals not to act on stereotypes or bias (e.g., accountability, affirmative action, etc.).
4. *Educating about bias processes*: Interventions primarily designed to create awareness, inform, or act as reminders (e.g., equality messages, diversity training).

Outcome Variables

I categorized outcome measures based on whether they assessed changes in workplace-relevant constructs related to stereotyping, prejudice, or discrimination. More specifically, outcomes fell into the following categories:

1. *Cognitive*: Measures assessing changes in beliefs, thoughts, opinions, or schemas about groups. Examples included perceptions of competence, skills for the job, and performance expectations for job candidates.
2. *Affective*: Measures assessing changes in emotions or feelings about groups. Examples included perceptions of warmth, friendliness, and the degree of liking for a job applicant.
3. *Behavior*: Measures assessing changes in actual behaviors, such as deciding whom to mentor, or behavioral intentions, such as willingness to hire.

Coding of Variables

First, the author and a trained assistant independently coded all relevant variables except for intervention category. To assess intercoder agreement, I compared the following data points: effect value, sample size, intervention type, sample type, and publication status. I calculated intercoder agreement as the percentage of agreement across these combined data points. From this analysis, 962 agreements and 78 disagreements were identified out of 1,040 data points (over 92% agreement for the variables of interest). Trained third and fourth coders independently coded the intervention category for each study, following the proposed intervention framework. Their coding was based on the intervention type (previously agreed by the first and second coders) and the descriptions of the treatment and control conditions. Figure 3 depicts the categorization of the 22 intervention types identified in this review based on this analysis. Interrater agreement for intervention category was calculated using Cohen's Kappa, with substantial agreement observed (κ = 92.8%, $p < .001$). Any discrepancies at each coding stage were discussed and resolved by the contributing coders by reviewing the studies in question. The hypotheses were unknown to all coders except for the author.

Analytical Approach

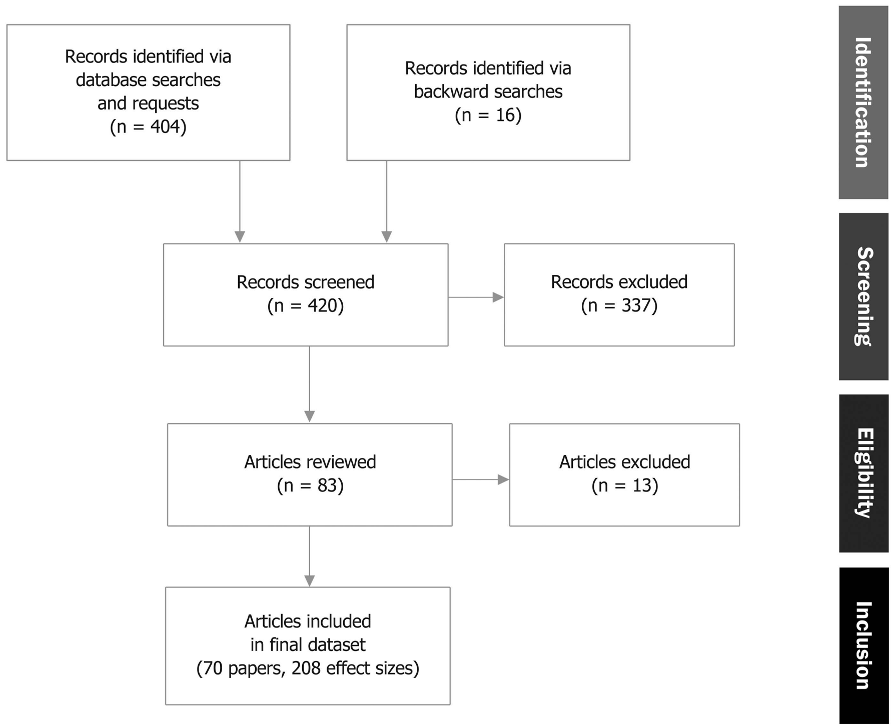
I computed the effects from the studies as the standardized mean difference (Cohen's d) between the intervention and control group outcomes and corrected to Hedges' g to adjust for bias in small samples (Hedges & Olkin, 1995). Additionally, I applied reliability corrections to adjust for statistical artifacts arising from measurement errors (Hunter & Schmidt, 2004). In cases where reliability estimates were unavailable, I adjusted values using the sample average reliability (Wiernik & Dahlke, 2020). If the requisite statistics (e.g., means and standard deviations) were unavailable, I calculated effect sizes by converting available statistics from the article or by directly contacting the first author (Borenstein et al., 2009; Higgins et al., 2019; Polanin & Snilstveit, 2016).

The meta-analytic procedures in this study followed the inverse-variance method, a common practice that places higher weights on studies providing more precise estimates by leveraging their sample size (Borenstein et al., 2009). Specifically, the weight assigned to each effect size was the inverse of its sampling variance ($w_i = 1/v_i$). Following convention, I assigned the sign of effect sizes such that studies with outcomes in the direction expected by the authors were coded as positive, while those with outcomes in the opposite

[1] While random assignment of participants into treatment and control groups was the preferred approach in the selection of interventional studies, I also included quasi-experimental field studies where random assignment was not feasible due to practical constraints (e.g., a district adopting anonymous applications).

Figure 2

Preferred Reporting Items for Systematic Reviews and Meta-Analyses Flow Diagram of Study Selections by Stage



Note. Flowchart depicting article search, review, selection, and output at different selection stages.

direction were coded as negative (Johnson & Eagly, 2000). Therefore, in this analysis, studies demonstrating a reduction in a biased outcome for the intervention group relative to the control group have positive effect sizes, while those showing an increase in biased outcomes have a negative size. Studies reporting null effects without additional statistics were coded as an effect size of zero (Paluck et al., 2021).

Several articles examined different intervention approaches across studies. For this reason, I included every effect size from an article that met the eligibility criteria. Given the nested structure of the data, I utilized multilevel models in every meta-analytical procedure to address potential dependencies in effect sizes derived from the same article. Multilevel meta-analytical models account for the possibility that effects within an article could be more similar to each other than across articles, and this approach results in unbiased estimates when compared to other meta-analytical procedures such as averaging effect sizes from the same article (Gooty et al., 2021; Moeyaert et al., 2017). Specifically, I employ two-level random effects meta-regression models in each analysis to account for the variance at the following levels: Level 1, representing within-article variance; Level 2, representing between-article variance (Van den Noortgate et al., 2013). Reflecting this nested structure at its most basic form, the two-level equations are as follows:

$$\text{Level 1: } Y_{ij} = \beta_{0j} + \varepsilon_{ij}, \quad \text{where } \varepsilon_{ij} \sim N(0, \sigma_{\varepsilon}^2), \tag{1}$$

$$\text{Level 2: } \beta_{0j} = \gamma_{00} + u_{0j}, \quad \text{where } u_{0j} \sim N(0, \sigma_{u}^2), \tag{2}$$

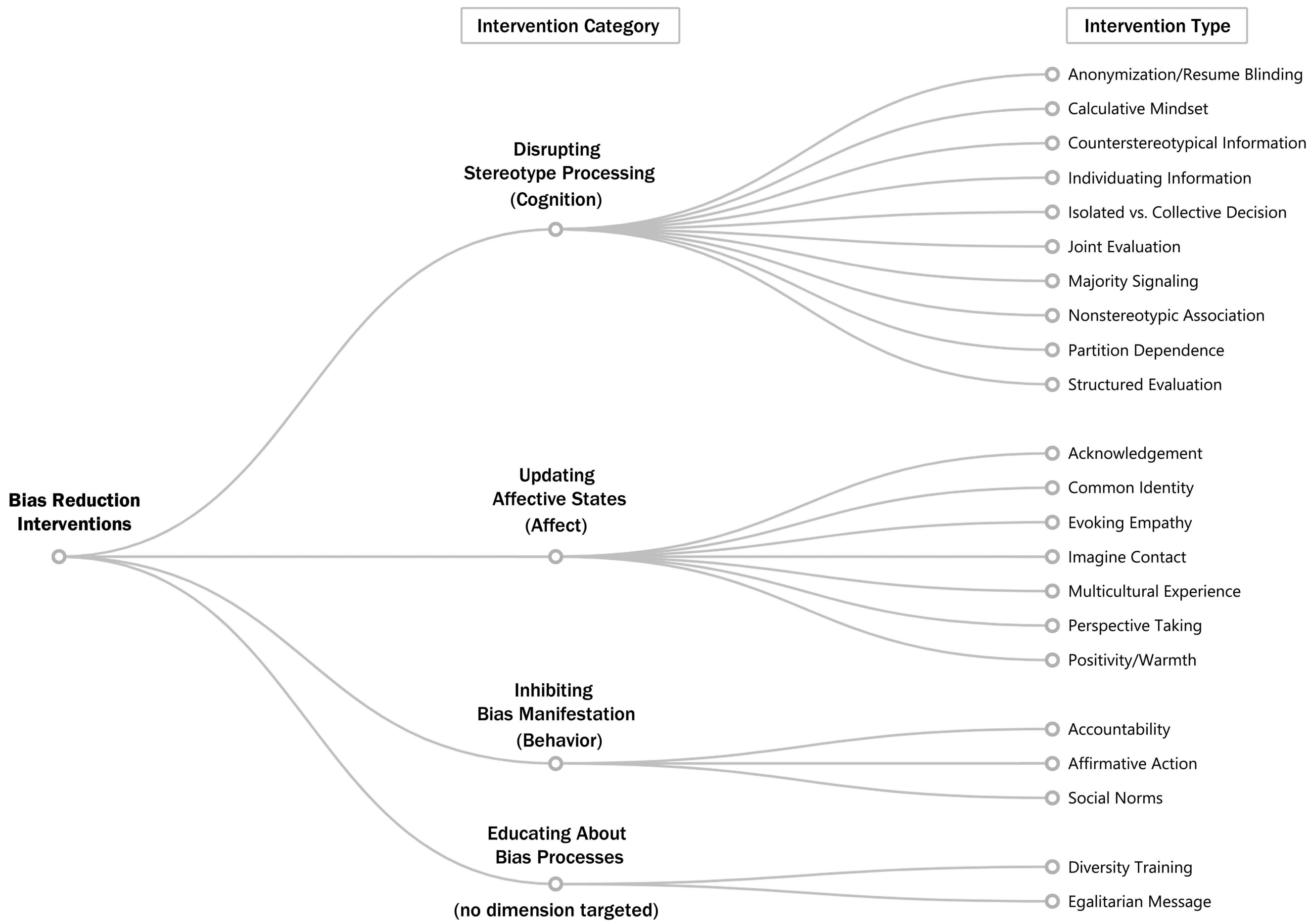
where Y_{ij} is the effect size for the ith outcome in the jth article, β_{0j} is the true effect size for the jth article, ε_{ij} is the residual error for the ith outcome in the jth article, γ_{00} is the overall intercept, and u_{0j} is the random effect at the article level.

To examine the potential for publication bias, I performed Egger's regression test by including the sampling variances of the effect sizes as a covariate in the model to accommodate for the nesting structure of the data (Fernández-Castilla et al., 2021). Because it is uncertain how common procedures to examine the robustness of meta-analytical results perform on multilevel models (e.g., funnel plots, trim-and-fill), I conduct sensitivity analyses on the existing data points in lieu of these methods. Following this approach, I reran the main analyses without influential outliers, identified via two methods: First, by excluding observations with Cook's distance values over $4/n$ (where n is the number of observations; $k = 2$ excluded), and second, by removing studies with hat values over twice the data set's average ($k = 16$ excluded; Cook, 1977; Fox, 2005).

Transparency and Openness

In conducting this examination, I adhered to the *Journal of Applied Psychology* methodological checklist for meta-analyses. In the previous sections, I detailed the inclusion criteria, coding procedures, and analytical plan. Preregistration, data, and code for this project are available at https://researchbox.org/1563. All analyses were conducted with the R software, Version 3.6.1

Figure 3
Taxonomy of Bias Reduction Interventions Chart



Note. Classification of bias reduction interventions by category based on the primary attitude dimension targeted by each intervention type.

(R Core Team, 2019), using the *metafor* package, Version 2.4-0 (Viechtbauer, 2010).

Results

Sample Characteristics

The final sample comprised 70 articles evaluating 22 intervention approaches based on 208 effect sizes. The number of effect sizes per article ranged from one to 12, with a median of two effect sizes per article. The studies examined the effects of workplace-related bias reduction interventions targeting various groups: racial minorities, LGBT+, gender, religion, pregnancy status, age, body weight, disability, physical attractiveness, and individuals from different nationalities. Cognitive outcome measures included implicit ($k = 3$) and explicit measures ($k = 48$), with most assessing changes in stereotyping levels (45.1%), followed by perceptions of competence (25.5%), fit/qualification for the job (17.6%), and expected performance (11.8%). Affective outcome measures mostly assessed perceptions of warmth or liking the target (59.1%) and prejudicial sentiment (18.2%). Behavioral outcome measures included formal ($k = 119$) and informal ($k = 16$) discrimination measures, with most assessing changes in hiring/selection decisions (63.7%) or contact measures (e.g., willingness to mentor or work with the target, 19.3%).

Nonhypothesized Results

I constructed meta-regression models in a stepwise manner to examine the associations between intervention categories, outcome variables, and intervention effectiveness.[2] The first analysis examined the presence of within- and between-article variance, consisting of a null model (i.e., intercept-only, Model 1) encompassing all 208 effect sizes. The results showed significant heterogeneity across effect sizes $Q(207) = 1025.26, p < .001$, suggesting the presence of moderators in the data set. When examining the result of the omnibus test, interventions appear to be modestly effective at reducing work-

[2] The author thank the input of an anonymous reviewer who assisted her in arriving at these analytical approaches and the action editor for their assistance in enhancing the methodological alignment with the theoretical foundations of this work. See Appendix B for specifications of these models.

related biased outcomes. On average, interventions tended to reduce all measures of workplace bias, compared to the control conditions, by approximately a fourth of a standard deviation, $g = 0.27$, $SE = 0.042$, $p < .001$, 95% CI [0.190, 0.356].

In Model 2, I examined the contribution of the outcome variable type in explaining the variability in effect sizes. To do so, I coded for the nature of the outcome measure (i.e., whether cognitive, affective, or behavioral outcomes). This model was not statistically significant, $Q_M(2) = 2.30$, $p = .317$, indicating that the type of outcome measure does not substantially contribute to explaining the variance in effect sizes across the studies. This result helps rule out the alternative explanation that certain types of outcomes merely produce larger effects than others.

In Model 3, I examine the contribution of intervention category in explaining variance in effect sizes. Here, I included the intervention category variable to Model 2 (0 = educating, 1 = disrupting, 2 = inhibiting, 3 = updating). This model was also not statistically significant, $Q_M(5) = 8.18$, $p = .146$. This result helps rule out the alternative explanation that certain intervention categories simply produce greater effects than others.

Hypothesized Results

To test the relationships in the conceptual framework, I estimated a model to investigate Hypotheses 1, 2, and 3. This model examines whether interventions targeting the behavioral, affective, and cognitive dimensions of attitudes produce better outcomes on measures of workplace discrimination, prejudice, and stereotyping when there is alignment between the targeted and the measured dimensions. I examined these effects by incorporating dummy variables for specific intervention category-outcome combinations. Specifically, I included dummy variables for the following arrangements: (a) "disruptive" category and cognitive outcome, (b) "updating" category and affective outcome, and (c) "inhibiting" category and behavioral outcome (1 = matching, 0 = not matching). The model yielded a statistically significant result, $Q_M(3) = 15.01$, $p = .002$. Examination of the estimates provided support for Hypotheses 1 and 3 but not for Hypothesis 2. Compared to interventions that did not match the attitudinal focus of the intervention with the outcome measure, matching these variables increased the effect size on the cognitive dimension by $g = 0.26$, $SE = 0.096$, $p = .007$, 95% CI [0.071, 0.449] and on the behavioral dimension by $g = 0.39$, $SE = 0.110$, $p < .001$, 95% CI [0.170, 0.601]. However, matching interventions and outcomes did not yield a reliable increase in the affective dimension, with a positive but not statistically significant increase in intervention effectiveness, $g = 0.07$, $SE = 0.101$, $p = .480$, 95% CI [−0.127, 0.269]. Overall, the results suggest an advantage in aligning the intervention category with the outcome measured. When looking at differences between a match between the intervention and outcome across attitude dimensions, a statistically significant result emerges $Q_M(1) = 10.39$, $p < .001$. Specifically, interventions not aligned with outcomes show a bias reduction of about $g = 0.24$, $SE = 0.041$. Meanwhile, aligning the focus of the intervention with the outcome variable increases the effect on average by $g = 0.19$, $SE = 0.059$, so moving from a small to a moderate effect size. Table 3 reports the relative effectiveness of intervention categories by outcome measure.

I next examined the fourth hypothesis, which states that educating about bias processes would be less effective in reducing bias measures relative to the other intervention categories. To investigate this, I estimated a model that differentiated between interventions focused on education about bias processes (coded as 1) and all other types of interventions (coded as 0). The findings indicate that educating about bias processes is notably less effective than other interventions, $Q_M(1) = 5.03$, $p = .025$. Specifically, the coefficient for the "educating" intervention category suggests a discernible reduction in effectiveness, equivalent to approximately one fifth of a standard deviation relative to all other intervention categories $g = -0.19$, $SE = 0.086$, 95% CI [−0.361, −0.024]. When segmenting results by outcome variable, the educational class of interventions failed to produce statistically significant reductions in any of the workplace bias measures, unlike the other intervention categories.

Table 3
Intervention Category Results by Outcome Type

Intervention category	k	Hedges' g	ρ	SE	p
Cognitive outcomes					
Disrupting stereotype processing	16	0.51***	0.57**	0.18	.002
Educating about bias processes	6	−0.20	0.35	0.19	.068
Inhibiting the manifestation of bias	5	−0.70**	−0.47	0.29	.104
Updating affective states	24	0.18	0.32*	0.14	.021
Affective outcomes					
Disrupting stereotype processing	4	0.62**	0.65***	0.18	<.001
Educating about bias processes	6	0.06	0.16	0.11	.159
Inhibiting the manifestation of bias	3	0.37	0.66***	0.17	<.001
Updating affective states	9	0.18	0.28**	0.10	.007
Behavioral outcomes					
Disrupting stereotype processing	72	0.22***	0.26***	0.04	<.001
Educating about bias processes	22	0.08	0.09	0.08	.219
Inhibiting the manifestation of bias	10	0.59***	0.64***	0.11	<.001
Updating affective states	31	0.23***	0.30***	0.06	<.001

Note. Meta-analytical results for interventions to reduce biased workplace outcomes by intervention category and outcome variable. ρ = Hedges' g values corrected for unreliability; k = number of effect sizes; SE = standard error.
* $p < .05$. ** $p < .01$. *** $p < .001$.

This result held whether the outcome was a cognitive, affective, or behavioral measure (p values between .068 and .219, see Table 3).

Publication Status

The Egger's regression test indicated the presence of substantial asymmetry in the data points ($p = .013$), which was also observed by significant moderation based on publication status $Q(1) = 8.43$, $p = .004$. In published articles, interventions achieved larger discrimination reduction effects on average $g = 0.32$, $SE = 0.044$ ($k = 182$) compared to unpublished articles $g = -0.004$, $SE = 0.103$ ($k = 26$). Based on sensitivity analyses, however, neither the hypothesized results nor the nonhypothesized models notably differed when analyzing the data without extreme outliers nor points of influential leverage (see Appendix C for results).

Discriminatory Outcomes

I next examined differences in the effectiveness of interventions when restricting the analyses to workplace behavioral outcomes only. Overall, intervention approaches appear modestly effective at reducing workplace discriminatory behavior. On average, interventions tended to reduce behavioral measures by approximately a fourth of a standard deviation, compared to a control condition, $g = 0.27$, $SE = 0.038$, $p < .001$, 95% CI [0.196, 0.344]. These results suggest that while the interventions were generally helpful in reducing workplace discriminatory outcomes, their impacts have been, on average, modest in magnitude.

Research Question 1: What Intervention Types Are Most Effective in Reducing Workplace Discrimination?

To examine which treatment approaches specifically contributed to the three significant intervention categories for discriminatory outcomes, I estimated a model at the intervention type level. Decomposing the intervention types into their different methods revealed 10 procedures that reached statistical significance. Figure 4 depicts summary effect sizes for all 22 intervention types for behavioral outcomes. In the disrupting stereotype processing category, leveraging structured evaluation $g = 0.51$, $SE = 0.115$, $p < .001$ and majority signaling by the targets (e.g., "whitening" resumes) $g = 0.44$, $SE = 0.184$, $p = .017$ produced the largest significant shifts in behavior. In the updating affective states category, imagining contact $g = 0.47$, $SE = 0.219$, $p = .031$ and multicultural experience $g = 0.46$, $SE = 0.212$, $p = .029$ caused the most substantial changes in behavioral measures. In the inhibiting bias manifestation category, making individuals accountable yielded the highest effect in reducing discrimination, $g = 0.80$, $SE = 0.185$, $p < .001$, followed by intervening on social norms, $g = 0.58$, $SE = 0.150$, $p < .001$.

Research Question 2: Are Interventions Targeting the Behavioral Dimension of Attitudes More Effective in Reducing Informal as Opposed to Formal Discriminatory Outcomes?

To examine this question, I fitted a model to estimate the separate effects for the two levels of the behavioral outcome, representing whether it captured informal or formal discrimination. Notably, interventions focused on the behavioral dimension of attitudes (i.e., inhibiting the manifestation of bias category) showed substantial reductions in both informal and formal discriminatory outcomes. At the same time, a nuanced examination reveals distinctive patterns in their effectiveness. Results indicate that interventions aimed at changing behavior produced larger effects on informal discrimination, $g = 0.49$, $SE = 0.093$, compared to formal discrimination measures, $g = 0.25$, $SE = 0.039$, $Q_M(1) = 6.81$, $p = .009$.

Discussion

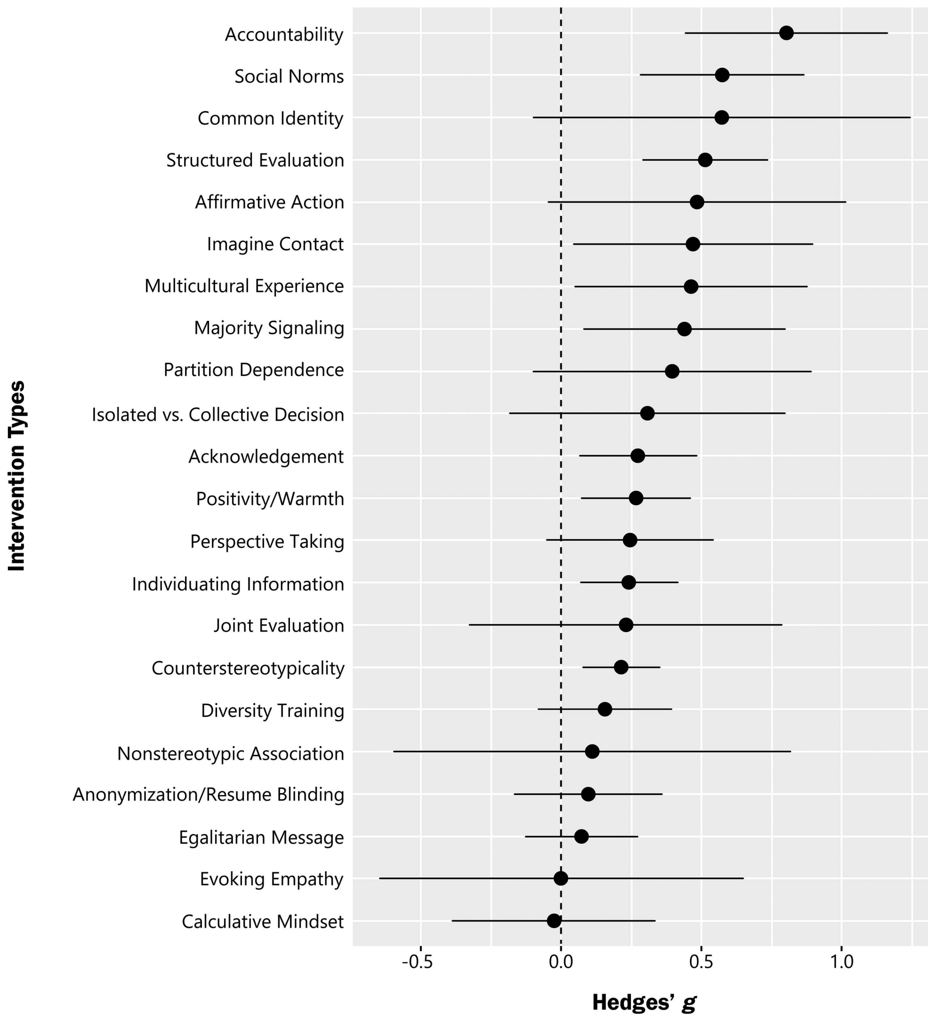
Despite extensive research aimed at reducing bias, knowing which interventions most effectively improve behavioral measures has been limited by the lack of a comprehensive taxonomy describing the attitude dimensions primarily targeted by different interventions. Drawing from the principle of compatibility and the tripartite attitude model, I proposed that interventions should be most effective in shifting bias measures when there is a correspondence between the attitude dimension being intervened upon and the desired outcome. A meta-analytical examination of studies assessing intervention-driven changes in discrimination, prejudice, and stereotyping suggests this model might help explain a possible reason for the differential effectiveness of workplace interventions.

While results demonstrated improved outcomes when cognitive and behavioral interventions matched their respective outcome dimensions, this pattern did not emerge for affective-based matches. The differential impact of intervention types and outcome matching on affective measures, as opposed to cognitive or behavioral measures, may stem from different factors. For instance, emotional responses are typically deeply ingrained, potentially making them more resistant to change than cognitive or behavioral measures (Edwards, 1990; Tropp & Pettigrew, 2005). Given this, it is possible that for affective variables, interventions targeting any of the attitude dimensions (cognitive, affective, or behavioral) would produce similar results, implying that dimension matching does not confer an enhanced effect for affective measures. Also, accurately gauging affective variables is notably challenging, potentially making them less reliable than cognitive or behavioral measures (Mauss & Robinson, 2009; Parrott & Hertel, 1999). Finally, affective measures may be more susceptible to social desirability bias and to individuals' lower introspective access to their emotional states, further complicating their measurement (Greenwald et al., 2009; Pearson et al., 2009).

Theoretical and Practical Implications

The present conceptual framework of bias interventions enables further theoretical advancement in this area by not only explaining which intervention category works best for behavioral outcomes but also *why*. In this regard, the results help reconcile some of the prior mixed findings in this literature. For example, Mousa (2020) found that an intergroup contact intervention (a typically affective-based treatment) between Christian and Muslim youth in postwar Iraq shifted behavioral measures while producing inconsistent or no attitudinal changes in the evaluative measures. In examining potential mechanisms, Mousa found evidence for a causal pathway via the newly introduced social norm, which normalized intergroup contact (i.e., a behaviorally focused treatment) but not via increased outgroup information (i.e., a cognitively based treatment), nor

Figure 4
Forest Plot by Intervention Type for Discriminatory Outcomes



Note. Forest plot synthesizing the effects of treatment versus control groups for intervention approaches measuring workplace-related behavioral outcomes with corresponding 95% confidence intervals.

increased empathy (i.e., an affectively based treatment)—a finding consistent with the model proposed here.

Another contribution stems from the finding that interventions focusing on the behavioral aspect of attitudes exhibit a higher impact on informal discrimination measures compared to formal discrimination. This observation, initially gleaned from field experiments indicating improvements in informal discrimination but not in formal discrimination outcomes from the same intervention, also aligns with the attitude dimension consistency perspective (see Morgan et al., 2013; Singletary & Hebl, 2009). In particular, the legal environment restricting discrimination creates more rigid norms in the workplace that are sufficiently strong to impact behavior relating to formal discrimination measures in the field (e.g., letting candidates complete an application). Informal discrimination variables lack such legal constraints, making them more malleable to interventions and allowing for more movement on these measures.

Focusing on reducing discriminatory behavior in the workplace, some of the least effective approaches based on this analysis bear mentioning—diversity training and equal opportunity messages—due to their ubiquity in organizations and anonymizing resumes—for the growing interest in its application (Agovino, 2020; Carlos, 2021). Short-term education about stereotypes and bias processes, as commonly done in corporate diversity training courses, and warning about equal opportunity policies prior to decisions did not yield promising results. These findings parallel other research demonstrating that antibias training does not generally reduce bias, alter behavior, or create workplace change (Dobbin & Kalev, 2018). The results on resume anonymization from this analysis also conform with wider research on the topic, which tends to find mixed results at best (Foley & Williamson, 2018). For instance, anonymized resumes generally show positive effects where discrimination is an issue in the first place but preclude more egalitarian approaches from companies already adopting a more pro-diversity attitude in their selection processes—as they impede recruiters from contextualizing ambiguous signals like employment hiatus and create a negative effect in such instances (Bertrand & Duflo, 2017).

Turning to the effective strategies that organizations can leverage in tackling discrimination, structured evaluations were effective in reducing biased behavior in the disrupting stereotype processing category. This aligns with research supporting the efficacy of this intervention approach across stigmatized identities (Bragger et al., 2002; Brecher et al., 2006; Levashina et al., 2014). Notably, the effect size for this intervention is similar in magnitude to the much less studied majority signaling intervention approach (g = 0.51 vs. g = 0.44). While studies show that minority job applicants who "whiten" their resumes by removing racial identifiers receive significantly more callbacks than identical racial-revealing resumes, this intervention approach is relatively underexamined, potentially due to its controversial nature. There is an opportunity to expand on unanswered questions about this intervention strategy, such as its generalizability, longer term effects past initial application screening, and psychological effects on targets (Arai & Skogman Thoursie, 2009; Kang et al., 2016). Also, while both majority signaling and anonymizing resumes obscure candidates' true demographic identity, majority signaling showed a significantly larger impact on reducing biased outcomes than resume blinding, with effect sizes of 0.44 and 0.10, respectively. A potential reason is that "whitening" resumes allows candidates to neutralize signals of nonprototypicality, whereas anonymization leaves this question open to the evaluator's interpretation.

In the updating affective states category, the most effective intervention was imagining contact. However, implementing this intervention in a consistent and nondisruptive way in the workplace context poses substantial challenges to its adoption. Finally, in the inhibiting bias manifestation category, accountability interventions proved most effective, followed by social norm treatments. Interestingly, research in social norm interventions to reduce workplace bias is currently sparse, leaving the specific features that might enhance or decrease its effectiveness unknown. Further research in this area is needed to understand questions such as the relative effectiveness of injunctive versus descriptive social norms and whether these are more effective if employed together rather than separately. Based on this review, accountability emerges as the most sensible approach for organizations to adopt in reducing discriminatory outcomes. Besides yielding the largest effect in changing behavior (g = 0.80), in practical terms, accountability can be more consistently deployed organization-wide relative to some of the other effective interventions. This involves making discretionary decisions contingent on accountability or as an integral factor in the decision-making process. For example, Castilla's (2015) field study demonstrated ways that organizations can employ accountability to facilitate fairer outcomes: by creating a committee overseeing pay decisions with the authority to modify choices deemed inadequate and by establishing a process where managers justify pay raises for each employee in their department.

Limitations and Future Research

Some limitations of this examination should be noted. First, due to the scarcity of studies measuring informal discrimination, I was unable to assess potential differences in effectiveness between intervention types along informal versus formal discrimination measures. As research that measures these variables simultaneously is currently limited, this presents a fertile area for further inquiry. Future studies could explore both measures concurrently to examine if differential outcomes emerge based on the intervention approach. Arguably, informal discrimination variables align more closely with measures of prejudice (e.g., hostility, making eye contact), so future research could examine whether these would be more responsive to interventions attempting to update affective states relative to inhibiting bias manifestation.

Second, while this study categorized interventions into broader groups, reflecting general shared characteristics to provide meaningful groupings for analyses, as noted by a reviewer, subclasses along these groups might warrant further exploration. For instance, cognitive measures could be further segmented into implicit versus explicit bias, providing a more nuanced understanding of these relationships. Previous research found no correlations between the effects of interventions on implicit and explicit bias, implying limits on when interventions attempting to modify one type of cognitive-based measure will produce changes in the other (Forscher et al., 2019; Paluck et al., 2021). Additionally, the positive or negative valence of emotions experienced could differentially influence the effectiveness of affective-based interventions (Dovidio et al., 2002). However, I did not examine these questions due to the insufficient number of studies in a particular condition or because they are beyond the scope of this study.

A recommendation for future workplace intervention research is to diversify discrimination outcomes beyond just candidate hiring or selection. For instance, formal discrimination measures can include promotions, stretch assignments, salary, and bonus allocations, and inverse measures such as who is laid-off or gets assigned "office housework"—low value-added tasks that do not advance one's career and are frequently asked of women of color (Tulshyan, 2018).

Conclusion

By synthesizing and systematically classifying bias reduction interventions, this analysis clarifies how organizations can improve their equity and fairness efforts. First, the present findings support a theorized model calling for the correspondence between outcome and the intervening attitude dimension to enhance intervention results. Practically, the results from this article suggest actionable steps organizations can take to reduce workplace discrimination. While the fruitfulness of some interventions may depend on an organization's status quo, accountability practices and structured evaluations appear as reasonable and practical strategies for researchers, businesses, and institutions to test and refine across organizational levels. Overall, further progress may be realized by understanding the relationships of these variables to improve interventional procedures and enable more diverse, equitable, and inclusive organizations.

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References marked with an asterisk indicate studies included in the meta-analyses.

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(*Appendices follow*)

Appendix A

Search Parameters

Database sources
APA PsycInfo, Google Scholar, Web of Science, Scopus, and ProQuest
Terms
work* OR job OR employ* OR hir* OR recruit* OR applicant OR promot* OR salary
AND
bias OR prejudic* OR discriminat* OR stereotyp*
AND
intervention OR treatment OR reduc* OR experiment*

Appendix B

Model Parameters

- Model 1 (null—intercept-only)

$$Y_{ij} + \gamma_{00} + u_{0j} + \varepsilon_{ij} \tag{B1}$$

- Model 2 (outcome variable)

$$Y_{ij} + \gamma_{00} + \gamma_{01}(\text{Outcomes}) + u_{0j} + \varepsilon_{ij} \tag{B2}$$

- Model 3 (intervention category variable)

$$Y_{ij} + \gamma_{00} + \gamma_{01}(\text{Outcomes}) + \gamma_{02}(\text{Intervention Category}) + u_{0j} + \varepsilon_{ij} \tag{B3}$$

- Model 4 (Hypotheses 1–3: matching intervention category and outcome variables = 1; not matching = 0)

$$Y_{ij} + \gamma_{00} + \gamma_{01}(\text{Cognitive Dummy}) + \gamma_{02}(\text{Affective Dummy}) + \gamma_{03}(\text{Behavior Dummy}) + u_{0j} + \varepsilon_{ij} \tag{B4}$$

- Model 5 (Hypothesis 4: "Educating" intervention category = 0; other intervention categories = 0)

$$Y_{ij} + \gamma_{00} + \gamma_{01}(\text{Educating Dummy}) + u_{0j} + \varepsilon_{ij} \tag{B5}$$

In these equations:

- Y_{ij} is the *i*th effect size in the *j*th article.
- γ_{00} is the overall intercept.
- $\gamma_{01:03}$ are coefficients for the respective variables.
- u_{0j} represents the random effect at the article level.
- ε_{ij} is the residual error at the effect size level.

(*Appendices continue*)

Appendix C

Sensitivity Analyses: Results Excluding Outliers

Variable	Q	Hedges' g	ρ	SE	p
Full sample (k = 208)					
Matching intervention and outcome	15.01				.002
Cognitive		0.24*	0.26**	0.10	.007
Affective		0.02	0.07	0.10	.480
Behavior		0.53***	0.39***	0.11	<.001
Influential points excluded based on Cook's distance (k = 206)					
Matching intervention and outcome	15.68				<.001
Cognitive		0.25*	0.26**	0.10	.006
Affective		0.02	0.07	0.10	.466
Behavior		0.53***	0.39**	0.11	<.001
Influential points excluded based on hat values (k = 192)					
Matching intervention and outcome	12.07				.007
Cognitive		0.21*	0.24*	0.10	.020
Affective		0.01	0.06	0.10	.537
Behavior		0.53***	0.37**	0.11	<.001

Note. Meta-analytical results for the hypothesized analyses with and without outliers and influential points of leverage. ρ = Hedges' g values corrected for unreliability. Cognitive: 1 = matches "disrupting stereotype processing" intervention category with outcome variables measuring cognitive dimensions of attitudes, 0 = no match between these variables. Affective: 1 = matches "updating affective states" intervention category with outcome variables measuring affective dimensions of attitudes, 0 = no match between these variables. Behavior: 1 = matches "inhibiting bias manifestation" intervention category with outcome variables measuring behavioral dimensions of attitudes, 0 = no match between these variables. SE = standard error.
* $p < .05$. ** $p < .01$. *** $p < .001$.

Appendix D

Exploratory Analyses on Discriminatory Outcomes

In exploratory analyses restricted to behavioral outcome variables, I examined whether differential effects emerged when accounting for some study characteristics or when adopting more granular classifications on intervention categories.

Moderator Variables

To examine potential moderation of the overall effect on workplace discriminatory outcomes, I recorded the following variables: (a) object of intervention (whether the treatment intervened on the target of evaluation, the evaluator, or the decision environment/architecture); (b) target group (coded for groups available in this sample: age, attractiveness, body weight, disability, gender, LGBT+, nationality, pregnancy, race or religion); (c) role (whether the job tested for an entry-level, professional, managerial, or faculty position)[D1]; (d) publication status (published or unpublished); and (e) sample type (convenience or field). Convenience samples included participants from online sources (e.g., Mechanical Turk, paid panels, project implicit pool) and undergraduate students. Field samples entailed surveying or observing individuals in natural settings, such as workers in retail stores, employees from multinational organizations, and university faculty members. Table D1 summarizes the meta-regression results fitted with these focal moderators. Results reveal no significant differences in intervention effectiveness for any of the variables examined.

Examining Subcategories for the "Disrupting Stereotype Processing" Category

I investigated whether the effectiveness of interventions targeting the cognitive dimension of attitudes varied when further classified based on whether they entirely prevented access to stereotypical information ("blocking" types) or simply disrupted its processing (nonblocking" types). I coded anonymizing/resume blinding and majority signaling as "blocking" intervention types, as they prevent exposure to demographic signals. The remaining intervention types within this category, such as structured evaluation, were coded as "nonblocking." No statistically significant differences emerged in reducing discriminatory outcomes when segmenting the disrupting intervention category this way ($g_{\text{blocking}} = 0.22$, $SE = 0.12$; $g_{\text{nonblocking}} = 0.27$, $SE = 0.06$, $p = .691$).

[D1] Nonfaculty job levels were classified as follows: entry-level (individual contributor role that does not typically require a 4-year degree or more specialized training, such as fast food operators, retail salesperson); professional (individual contributor role that generally requires either a 4-year degree or more specialized training, such as software engineer, marketing analyst); and managerial (title was indicative of supervisory responsibilities, such as department head, bank branch manager, etc.).

(*Appendices continue*)

Table D1
Moderator Analysis Results

Variable	Q	k	Hedges' g	ρ	SE	p
Intervention object	0.78					.678
Architecture		14	0.03	0.17	0.13	.175
Evaluator		127	0.26***	0.29***	0.05	<.001
Target		67	0.22**	0.26***	0.07	<.001
Target group	8.13					.522
Age		11	0.15	0.16	0.18	.375
Body weight		20	0.11	0.20	0.12	.090
Disability		17	0.01	0.10	0.16	.550
Gender		58	0.23***	0.28***	0.06	<.001
LGBT+		17	0.38**	0.41**	0.14	.003
Nationality		33	0.26***	0.29***	0.07	<.001
Physical attractiveness		6	0.79**	0.87**	0.28	.002
Pregnancy		16	0.14	0.15	0.17	.371
Race		24	0.23**	0.27***	0.07	<.001
Religion		6	0.10	0.11	0.28	.686
Role	1.25					.870
Entry-level		45	0.25**	0.31***	0.09	<.001
Faculty		3	0.10	0.36*	0.17	.033
Managerial		28	0.25*	0.12	0.11	.265
Professional		54	0.24***	0.27***	0.08	<.001
Sample type	3.75					.053
Convenience		132	0.27***	0.30***	0.05	<.001
Field		76	0.17**	0.22***	0.06	<.001

Note. Results for multilevel meta-regressions based on moderator analyses. ρ = Hedges' g values corrected for unreliability; k = number of effect sizes contributing to estimate calculations for each level. Positive ρ values indicate a reduction of discrimination in the intervention group relative to the control group. SE = standard error.
$^{*}p < .05$. $^{**}p < .01$. $^{***}p < .001$.

Examining Subcategories for the "Inhibiting Bias Manifestation" Category

Within interventions targeting the behavioral dimension, I examined whether the level of discretion exercised by an evaluator impacted effectiveness in reducing discrimination. I classified interventions that allowed for more discretion, where the decision-maker could rely more on normative information or their own commitments, as "discretionary" intervention types. In contrast, "less discretionary" interventions tend to create more programmed decisions due to their more structured nature. Within this intervention category, I coded the following intervention types as "discretionary": social norms and accountability. I coded the intervention type "affirmative action" as "less discretionary." No statistically significant differences emerged in reducing discriminatory outcomes when segmenting the inhibiting intervention category based on the level of discretion ($g_{\text{discretionary}} = 0.71$, $SE = 0.10$; $g_{\text{less-discretionary}} = 0.49$, $SE = 0.17$; $p = .257$).

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